

A DNA assay for rapid discrimination between beaver species as a tool for alien species management

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Abstract The confirmed presence of alien North American beavers in some regions of Eurasia may compete with and hinder the successful recolonisation of the native Eurasian species back to its former range. Distinguishing the two species in the field can be problematic, time consuming and expensive, thereby potentially limiting appropriate conservation actions. Here, a rapid and inexpensive genetic SNP assay is described that can separate the two species from either non-invasively collected samples or samples taken directly from restrained individuals. We applied these new genetic assays to free-living beavers of unknown origin sampled in Scotland.

Keywords *Castor fiber* · *Castor canadensis* · Real-time PCR · SNP · Wildlife management · Non-invasive sampling

Introduction

Two extant species of beaver are now recognised through a difference in chromosome number (Eurasian *Castor fiber* = 48, North American *Castor canadensis* = 40) (Lavrov and Orlov 1973); these diverged approximately 7.5 million years ago (Horn et al. 2011). However, Eurasian (*Cf*) and North American (*Cc*) beavers are morphologically, behaviourally and ecologically very similar, making distinguishing these species difficult in the field (Rosell et al. 2005; Danilov et al. 2011a, 2011b). Current techniques for species identification include the examination of the colour and viscosity of anal gland secretions (AGS) in correctly sexed beavers (Rosell and Sun 1999) and/or genetic analysis (Kuehn et al. 2000; Dewas et al. 2012). Although the 2000 examination of AGS is a practical and quick field method, it does require animal capture, experienced sample collection and may be problematic if an animal is sexed incorrectly, which is more likely in younger animals (Osborn 1955; Rosell and Sun 1999). While genetic tests are more definitive and can allow for species identification without catching or disturbing the animal through the use of non-invasive hair trapping (Herr and Schley 2009), such methods still tend to be time consuming and expensive, as they require the generation and subsequent analysis of DNA sequence information.

Both species of beaver were almost brought to extinction largely as a result of fur hunting, with *Cf* reduced to ~1,200 individuals scattered across 8 isolated populations across Europe by the late nineteenth century (Halley and Rosell 2002). Since the second half of the twentieth century, various conservation strategies, including hunting regulations, reintroduction and translocation programs, along with natural spread have seen both species successfully recover and recolonise large parts of their former native range. Prior to the determination of two separate species, seven individuals of *Cc* were imported to Finland to supplement a reintroduction program

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of *Cf* from Norway (Lahti and Helminen 1974). In the absence of competition from *Cf*, significant predation, including hunting pressure, and with appropriate environmental factors, their numbers rose rapidly and were further translocated to other parts of Finland (Parker et al. 2012). Free-living *Cc* is now reported at a number of locations across Europe, including additional escapees from zoological collections (Michaux et al. 2011; Dewas et al. 2012). Although Lavrov (1996) reported a post-zygotic early stage foetus hybrid between the two species, previous and subsequent active attempts in captivity to breed hybrids have not resulted in viable offspring (Djoshkin and Safonow 1972; Kuehn et al. 2000; Lavrov and Orlov 1973; Zurowski 1983), and therefore hybridisation between the two species is considered unlikely. The eradication of *Cc* from Europe is a viable and legally supported conservation strategy but given the protected status of *Cf* presents practical problems for wildlife management personnel. Expanding populations of both species are converging at two main fronts, in Finland and Northwest Russia (Parker et al. 2012); therefore, positive identification of *Cc* individuals followed by dispatch or sterilisation ensures complete removal of this alien species if resources, public and political are obtainable (Parker et al. 2012).

From a practical and welfare perspective, the wide-scale trapping and holding of beavers for species identification is unrealistic. Genetic identification of species is preferable, particularly when translocating or reintroducing animals. Therefore, an effective, robust test providing quick results whilst an individual animal is restrained could offer a practical solution which could be deployed within key areas to facilitate the removal of this alien species. Moreover, the use of non-invasive hair trapping promises to effectively screen for the presence of *Cc* without animal trapping and restraint (Herr and Schley 2009).

Method

DNA samples collected from tissue, blood and hair samples were collected from both beaver species across their species range. In addition, DNA samples from other mammal species with overlapping niche/distributions were also incorporated to assess the species specificity of any assay (Table 1). In addition, four beaver samples were provided from unknown species found shot, road victim or lawfully trapped in an area of Scotland where beavers of unknown provenance have been accidentally or wilfully released.

DNA was extracted using the QIAGEN Investigator kits following the supplied protocols. All DNA extracts were normalised to 6.5 ng/μl using a NanoDrop spectrophotometer. From a consensus of six complete mitochondrial genomes, five *Cf*, representing the *albicus*, *birulai*, *tuvanicus*, *pohlei* and *belorussicus* subspecies (Durka et al. 2005) and a single *Cc*

(from a introduced population in Eastern Scandinavia) (Horn et al. 2011), two single nucleotide polymorphisms (SNPs; see, e.g. Morin et al. 2004) were identified in the 16S rRNA gene that were both diagnostic and suitable for the design of a SNP-based assay, i.e. conserved flanking region. The first SNP assay targeted was a T/C (*Cc/Cf*) mutation at position 1971, and the second targeted a G/A (*Cc/Cf*) mutation at position 2473 (Horn et al. 2011). An inexpensive ‘allele-specific primer’ based assay, KASPar probe (KBiosciences), was designed for both SNP positions labelling the two SNP alleles for *Cc* (T,G) with a FAM fluorophore and for the *Cf* (C,A) alleles with a HEX fluorophore. For SNP position 1971, the common reverse primer was 5′ TGGGCAGGCGGTGCCTCTAATA 3′ and the two forwards FAM 5′ CCAAAAACATCACCTCTAGCATTACAAT 3′ and HEX 5′ CAAAAACATCACCTCTAGCATTACAAC 3′. For SNP position 2473, the common reverse primer was 5′ CGATGTTGGATCAGGACATCCCAAT 3′ and the two forwards FAM 5′ GAACAAACGAACCTTTAATAGCGGC 3′ and HEX 5′ GAACAAACGAACCTTTAATAGCGGT 3′.

Species identification could then be determined by preferential incorporation of either the HEX or FAM fluorophore during amplification for each SNP. PCR amplifications for both SNPs were carried out individually but simultaneously in a single PCR amplification using an Applied Biosystems StepOne real-time thermal cycler and following the standard amplification conditions for KASPar probes as recommended by the manufacturer. Reaction volumes of 8 μl were used and consisted of 5 μl of real-time PCR mastermix (KBioscience) containing ROX as an internal normalising control, 3 μl of normalised DNA and 0.1 μl of the SNP primer mix. All amplifications incorporated two ‘no template controls’. SNP allele designations were called automatically using a two cluster algorithm as heterozygotes are not present, based on a 95 % confidence cluster calling using the proprietary StepOne analysis software. For the unknown beaver species samples and the additional mammal species selected for specificity testing, real-time PCR was undertaken for both SNP positions that incorporated two positive controls for each species and two ‘no template controls’.

Results and discussion

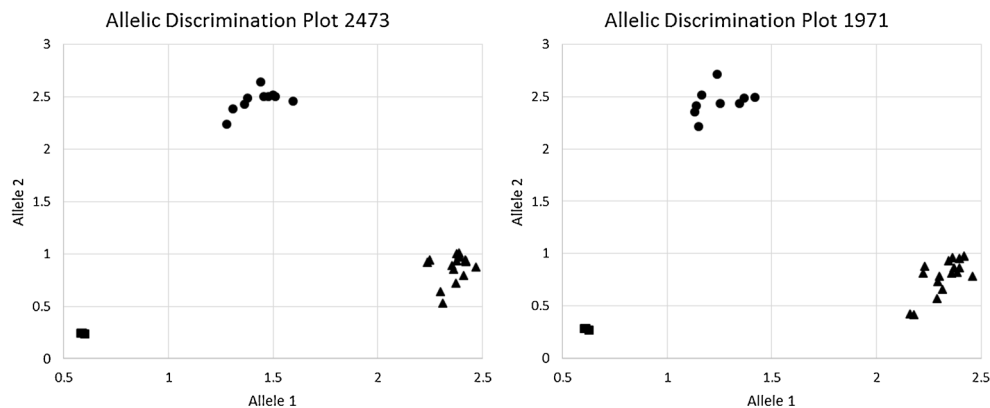
The results from our *Cf* subspecies samples and *Cc* samples (Table 1) concur with previous studies from other source locations (Horn et al. 2011) revealed that SNP positions 1971 and 2473 in the 16-s mitochondrial gene are fixed for nucleotides C/A, respectively, in *Cf* and G/T, respectively, in *Cc*, thereby making these positions ideal for species identification purposes. Both SNP assays developed here unambiguously identified the correct species in all authenticated samples of North American and Eurasian beaver (Fig. 1). This

Table 1 Mammal DNA samples used for identification and discriminations of beaver species

Species	Source	Sample type	n	snp 1971 Amplification efficiency /genotype	snp 2473 Amplification efficiency/genotype
<i>Castor fiber</i>	<i>C.f. fiber</i>	Tissue	8	*** /T	*** /A
		Germany	8	*** /T	*** /A
			1	*** /T	*** /A
		Hair	5	*** /T	*** /A
	<i>C.f. galliae</i>	Hair	5	*** /T	*** /A
		<i>C.f. albicus</i>	5	*** /T	*** /A
			1	*** /T	*** /A
	<i>C.f. ?</i>	Hair	1	*** /T	*** /A
<i>Castor canadensis</i>	Germany	Tissue	2	*** /C	*** /G
		USA	7	*** /C	*** /G
				*** /C	*** /G
<i>Castor sp. ?</i>	Scotland	Tissue	4	*** /T	*** /A
<i>Bos taurus</i>	Germany	Hair	1	*	*
<i>Canis lupus</i>	Germany	Scat	1	*	** /A
<i>Canis lupus familiaris</i>	Germany	Hair	1	*	*
<i>Capra aegagrus hircus</i>	Zoo/Germany	Hair	1	*	*
<i>Cervus elaphus</i>	Zoo/Germany	Hair	1	*	*
<i>Capreolus capreolus</i>			1	*	*
<i>Dama dama</i>	Zoo/Germany	Hair	1	*	*
<i>Equus ferus caballus</i>	Germany	Scat	1	*	*
<i>Felis silvestris</i>	Germany	Hair	1	*	*
<i>Felis silvestris catus</i>	Germany	Hair	1	*	*
<i>Glis glis</i>	Germany	Tissue	1	*	** /G
<i>Homo sapiens</i>	Germany	Hair	1	*	*
<i>Lepus europaeus</i>	Germany	Hair	1	*	** /A
<i>Lynx lynx</i>	Zoo/Germany	Hair	1	*	*
<i>Martes foina</i>	Germany	Hair	1	*	*
<i>Meles meles</i>	Bulgaria	Hair	1	*	*
<i>Oryctolagus cuniculus</i>	Germany	Hair	1	*	** /G
<i>Ovis orientalis aries</i>	Zoo/Germany	Scat	1	*	*
<i>Procyon lotor</i>	Germany	Hair	1	*	** /G
<i>Sus scrofa</i>	Germany	Tissue	1	*	*
<i>Ursus arctos</i>	Bulgaria	Tissue	1	*	*

Amplification efficiency *** strong amplification, ** some amplification, * no amplification

Fig. 1 Results of allele-specific real-time PCR amplification. Cc (FAM labelled) primers amplify in the vertical direction (circles), Cf (HEX labelled) primers amplify in the horizontal position (triangles). Back squares are no template controls



result demonstrates that this technique is a useful method for discrimination between these species. The four unknown beaver samples clustered for both SNPs with the Eurasian beaver identifying them as *Cf* (data not shown), the correct species for this geographical distribution. SNP 1971 failed to amplify in any other mammal species tested; however, SNP 2473 showed non-specific amplification in several other mammals (Table 1). Therefore, either, and preferably both, SNP markers can be utilised for known beaver samples to ascertain species, while for non-invasively collected samples, such as hair traps or faeces, SNP 1971 should be utilised to avoid false-positive results in terms of presence or species of beaver.

The presence of North American beaver in some regions may not only hinder the successful recolonisation of the Eurasian beaver back to its former range, but also presents a significant animal management issue (see Parker et al. 2012). The DNA identification assay described here offers a robust and rapid method for determining the species of beaver when this is unknown regardless of the subspecies or population of *Cf*. By utilising a real-time PCR technique, a species identification result can be provided in less than 3 h from delivery of a blood or hair sample from a trapped animal, a substantial improvement compared to Sanger sequencing of mitochondrial genes. Importantly, this technique is also extremely inexpensive with reagent costs including DNA extraction being approximately € 3.5 per sample analysed, considerably less than standard Sanger DNA sequencing and more amenable to testing single or low numbers of samples more typical of conservation or management projects.

As genomic information becomes more widely available and as techniques improve and costs fall, DNA testing has the potential to be used much more widely as an applied ecological and conservation management tool (e.g. McEwing et al. 2011).

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